

MATH 227C: Mathematical Biology

Homework 6

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Problem 1

Consider the chemical reaction system $A \rightarrow 2A$, $A \rightarrow \phi$, with rate parameters c_1 and c_2 respectively.

- a) Write down the master equation for this system, using the variable $\pi_i(t) = P(A(t) = i)$.
- b) Calculate explicitly the ODE for $E(X(t))$. Compare it with that for the deterministic ODE of the system
- c) Implement a few simulations for $c_1 = 10$ and either $c_2 = 9.5$ or $c_2 = 10.5$.
- d) Do you expect the master equation to have a stationary distribution? Describe the long term behavior for either case $c_1 < c_2$ and $c_1 > c_2$.

Let $\pi_i(t) = P(X(t) = i)$, where $X(t)$ denotes the number of A -molecules at time t . The reaction system is



For a state with i molecules, the reaction $A \rightarrow 2A$ increases the count by 1, while $A \rightarrow \phi$ decreases it by 1. The total jump rate out of state i is therefore $(c_1 + c_2)i$.

Master equation

For $i \geq 1$, the probability $\pi_i(t)$ changes because the process can jump into state i from states $i - 1$ and $i + 1$, and it can also jump out of state i . Thus,

$$\frac{d\pi_i(t)}{dt} = c_1(i-1)\pi_{i-1}(t) + c_2(i+1)\pi_{i+1}(t) - (c_1 + c_2)i\pi_i(t), \quad i \geq 1.$$

For the boundary state $i = 0$, only the reaction $A \rightarrow \phi$ can enter state 0 from state 1. Hence,

$$\frac{d\pi_0(t)}{dt} = c_2 \pi_1(t).$$

Putting these together, the master equation is

$$\frac{d\pi_i(t)}{dt} = c_1(i-1)\pi_{i-1}(t) + c_2(i+1)\pi_{i+1}(t) - (c_1 + c_2)i\pi_i(t), \quad i \geq 1,$$

with

$$\frac{d\pi_0(t)}{dt} = c_2 \pi_1(t).$$

Equation for $E[X(t)]$

Let

$$m(t) = E[X(t)] = \sum_{i=0}^{\infty} i \pi_i(t).$$

Differentiate both sides with respect to t :

$$\frac{d}{dt}m(t) = \sum_{i=0}^{\infty} i \frac{d\pi_i(t)}{dt}.$$

Using the master equation for $i \geq 1$,

$$\frac{d\pi_i(t)}{dt} = c_1(i-1)\pi_{i-1}(t) + c_2(i+1)\pi_{i+1}(t) - (c_1 + c_2)i\pi_i(t),$$

we obtain

$$\frac{d}{dt}m(t) = \sum_{i=1}^{\infty} i \left[c_1(i-1)\pi_{i-1}(t) + c_2(i+1)\pi_{i+1}(t) - (c_1 + c_2)i\pi_i(t) \right].$$

Now split the sum into three parts:

$$\frac{d}{dt}m(t) = c_1 \sum_{i=1}^{\infty} i(i-1)\pi_{i-1}(t) + c_2 \sum_{i=1}^{\infty} i(i+1)\pi_{i+1}(t) - (c_1 + c_2) \sum_{i=1}^{\infty} i^2\pi_i(t).$$

To simplify the first sum, set $j = i - 1$. Then $i = j + 1$, and

$$\sum_{i=1}^{\infty} i(i-1)\pi_{i-1}(t) = \sum_{j=0}^{\infty} (j+1)j\pi_j(t).$$

For the second sum, set $j = i + 1$. Then $i = j - 1$, and

$$\sum_{i=1}^{\infty} i(i+1)\pi_{i+1}(t) = \sum_{j=2}^{\infty} (j-1)j\pi_j(t).$$

Therefore,

$$\frac{d}{dt}m(t) = c_1 \sum_{j=0}^{\infty} (j+1)j \pi_j(t) + c_2 \sum_{j=2}^{\infty} (j-1)j \pi_j(t) - (c_1 + c_2) \sum_{j=0}^{\infty} j^2 \pi_j(t).$$

Now expand the factors:

$$(j+1)j = j^2 + j, \quad (j-1)j = j^2 - j.$$

So

$$\frac{d}{dt}m(t) = c_1 \sum_{j=0}^{\infty} (j^2 + j) \pi_j(t) + c_2 \sum_{j=0}^{\infty} (j^2 - j) \pi_j(t) - (c_1 + c_2) \sum_{j=0}^{\infty} j^2 \pi_j(t).$$

The j^2 -terms cancel, leaving

$$\frac{d}{dt}m(t) = (c_1 - c_2) \sum_{j=0}^{\infty} j \pi_j(t).$$

Since $\sum_{j=0}^{\infty} j \pi_j(t) = E[X(t)]$, we get

$$\frac{d}{dt}E[X(t)] = (c_1 - c_2) E[X(t)].$$

Hence,

$$E[X(t)] = E[X(0)] e^{(c_1 - c_2)t}.$$

Comparison with the deterministic model

The deterministic rate equation is

$$\frac{dx}{dt} = (c_1 - c_2)x.$$

Thus the expected value of the stochastic model satisfies the same equation as the deterministic system.

Simulations and Long-Term Behavior

Figure 1 shows a few sample paths for the stochastic birth-death process with $c_1 = 10$ and two choices of c_2 : $c_2 = 9.5$ and $c_2 = 10.5$. Figure 1 shows the corresponding deterministic mean-field solutions. The two cases behave very differently because the sign of $c_1 - c_2$ determines whether the process has a net upward or downward drift.

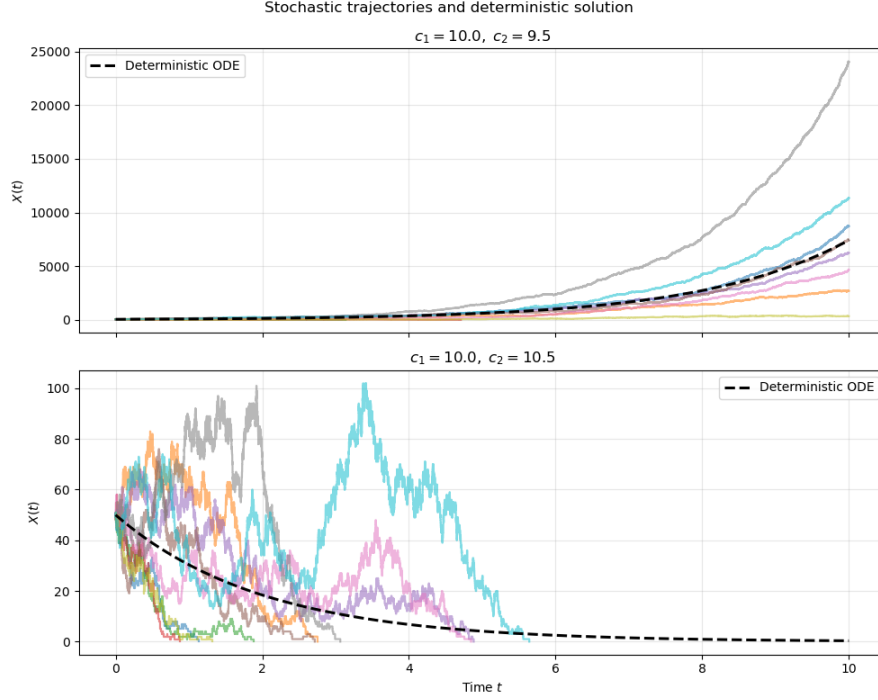


Figure 1: Sample stochastic trajectories for the birth-death process with $c_1 = 10$. The upper panel corresponds to $c_2 = 9.5$, and the lower panel corresponds to $c_2 = 10.5$. Deterministic solutions shown as dashed lines.

Case 1: $c_1 > c_2$

When $c_1 = 10$ and $c_2 = 9.5$, births occur slightly more often than deaths. The deterministic equation

$$\frac{dx}{dt} = (10 - 9.5)x = 0.5x$$

has exponentially growing solutions. This is reflected in Figure 1, where the trajectory increases with time. For the stochastic model, the sample paths in Figure 1 also tend to move upward, although they fluctuate randomly around that trend. Because the mean population grows, the master equation does *not* have a stationary distribution supported on finite states in this supercritical case. The probability mass drifts toward larger and larger values.

Case 2: $c_1 < c_2$

When $c_1 = 10$ and $c_2 = 10.5$, deaths occur more often than births. The deterministic equation becomes

$$\frac{dx}{dt} = (10 - 10.5)x = -0.5x,$$

so the mean decays exponentially to zero. Figure 1 shows this downward trend clearly, and the stochastic paths in Figure 1 typically move toward extinction as well. In this subcritical case, the process is pulled toward the absorbing state $X = 0$. Hence the long-term behavior is extinction with probability one, and the stationary distribution is the degenerate distribution concentrated at $X = 0$.

In summary, the master equation has **no nontrivial stationary distribution for $c_1 > c_2$** , because the population keeps growing. For $c_1 < c_2$, the process eventually dies out, and the only **stationary distribution is the trivial one at zero**.

Problem 2

Suppose a coin has a probability θ of heads, where θ is unknown. An experiment involving a total of 100 coin throws leads to 75 heads and 25 tails.

- a) Implement a Bayesian algorithm to determine the posterior distribution $\pi(\theta|X = x)$ of parameter values. Assume as a prior distribution $\pi(\theta)$ the uniform distribution.
- b) Plot the posterior distribution (give or take a constant multiple) as a function of θ . What is a reasonable range for the value of θ ?

Let θ be the probability of heads, and suppose we observe $x = 75$ heads in $n = 100$ tosses. We assume a uniform prior on θ , so

$$\pi(\theta) = 1, \quad 0 \leq \theta \leq 1.$$

Bayesian update

For 75 heads and 25 tails, the likelihood is binomial:

$$L(\theta; x) = P(X = x | \theta) \propto \theta^{75}(1 - \theta)^{25}.$$

By Bayes' formula,

$$\pi(\theta | X = x) \propto L(\theta; x) \pi(\theta).$$

Since the prior is uniform, this gives

$$\pi(\theta | X = x) \propto \theta^{75}(1 - \theta)^{25}.$$

This is the kernel of a Beta distribution. Therefore,

$$\theta \mid X = x \sim \text{Beta}(76, 26).$$

So the posterior density is

$$\pi(\theta \mid X = x) = \frac{1}{B(76, 26)} \theta^{75} (1 - \theta)^{25}, \quad 0 \leq \theta \leq 1.$$

Plot and interpretation

To generate the plot, the posterior density was first written in unnormalized form as

$$\pi(\theta \mid X = x) \propto \theta^{75} (1 - \theta)^{25},$$

which follows from the binomial likelihood with a uniform prior on θ . This expression was then evaluated on a fine grid of $\theta \in [0, 1]$, normalized numerically so it could be drawn as a density curve, and plotted with the result shown in Figure 2. A vertical marker for the posterior mean and a shaded credible interval can also be included to make the concentration of probability easier to see.

The posterior density, up to a constant factor, is shown in Figure 2. It is strongly concentrated near $\theta = 0.75$, which matches the observed proportion 75/100.

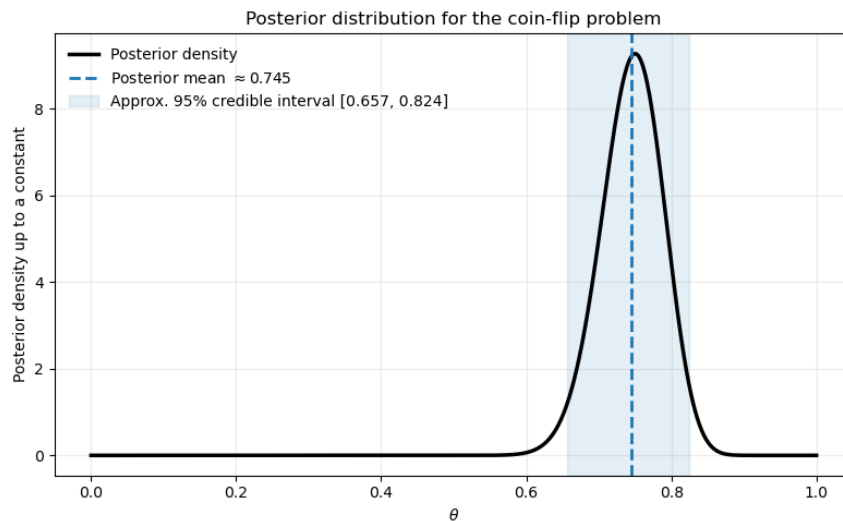


Figure 2: Posterior distribution for θ given 75 heads in 100 coin tosses, using a uniform prior.

A reasonable range for θ is the region where the posterior has most of its

mass. From the plot, a sensible interval is roughly

$$0.66 \lesssim \theta \lesssim 0.82.$$

In particular, the posterior mean is

$$E[\theta \mid X = x] = \frac{76}{76 + 26} \approx 0.745.$$

So the data suggest that the coin is biased toward heads, with θ close to 0.75.

Therefore, using a uniform prior, the posterior is $\text{Beta}(76, 26)$. The most reasonable values of θ are around 0.75, with most of the posterior mass lying roughly between 0.66 and 0.82.

Problem 3

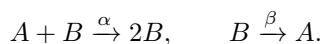
Consider the reaction $A + B \rightarrow 2B$, $B \rightarrow A$, with parameters α and β respectively. Notice the mass conservation $A + B = n$

- a) Write down the master equation of this system using the notation $\pi_j(t) = P(A(t) = j, B(t) = n - j)$.
- b) Write down the Markov chain for the stochastic system assuming $A + B = n$ for fixed n . Notice what happens when $A = n, B = 0$ in the Markov chain.
- c) What do you expect will be the long term behavior of this system given your answer in b)? Write the stationary distribution π .
- d) Implement stochastic simulations for $n=10$ and $n=100$. Compare these simulations with simulations of the deterministic ODE for this system.

Let $\pi_j(t) = P(A(t) = j, B(t) = n - j)$, where $j = 0, 1, \dots, n$. Since $A + B = n$ is conserved, the system can be described by the single variable $A(t)$, and $B(t) = n - A(t)$.

Master equation

There are two reactions:



If the system is in state j , then $A = j$ and $B = n - j$.

The reaction $A + B \rightarrow 2B$ decreases A by one, so it sends state j to $j - 1$. Its rate in state j is

$$\alpha A B = \alpha j(n - j).$$

The reaction $B \rightarrow A$ increases A by one, so it sends state j to $j + 1$. Its rate in state j is

$$\beta B = \beta(n - j).$$

Therefore, for $1 \leq j \leq n - 1$, the master equation is

$$\frac{d\pi_j(t)}{dt} = \alpha(j+1)(n-j-1)\pi_{j+1}(t) + \beta(n-j+1)\pi_{j-1}(t) - [\alpha j(n-j) + \beta(n-j)]\pi_j(t).$$

At the boundary states, the same logic gives

$$\frac{d\pi_0(t)}{dt} = \alpha(1)(n-1)\pi_1(t) - \beta n \pi_0(t),$$

and

$$\frac{d\pi_n(t)}{dt} = \beta \pi_{n-1}(t),$$

because the reaction $A + B \rightarrow 2B$ is impossible when $B = 0$, and the reaction $B \rightarrow A$ is impossible when $B = 0$ as well.

Markov chain

The stochastic dynamics form a continuous-time Markov chain on the state space

$$\{0, 1, 2, \dots, n\},$$

where state j means $A = j$ and $B = n - j$. From state j , the process can move:

$$j \rightarrow j - 1 \quad \text{with rate } \alpha j(n - j),$$

$$j \rightarrow j + 1 \quad \text{with rate } \beta(n - j).$$

So the chain is a birth-death process on the finite set $\{0, \dots, n\}$.

When $A = n$, $B = 0$, that is, at state $j = n$, both transition rates vanish:

$$\alpha n(n - n) = 0, \quad \beta(n - n) = 0.$$

Thus $j = n$ is an absorbing state. Once the system reaches $A = n$, it stays there forever.

Long-term behavior and stationary distribution

Because $j = n$ is absorbing, we expect the process to drift eventually to the state $A = n$, $B = 0$. Indeed, from any state with $j < n$, there is a positive rate of moving upward toward larger A , while the absorbing state cannot be left once reached. So the **long-term behavior is extinction of B and fixation at $A = n$** .

Therefore the stationary distribution is the point mass at the absorbing

state:

$$\pi_j = \begin{cases} 1, & j = n, \\ 0, & j = 0, 1, \dots, n-1. \end{cases}$$

In vector form, this is

$$\pi = (0, 0, \dots, 0, 1).$$

Thus the only stationary distribution is concentrated at $A = n, B = 0$.

Stochastic simulations and deterministic comparison

Since $A(t) + B(t) = n$, it is enough to track the variable $A(t)$. If we define the normalized quantity

$$x(t) = \frac{A(t)}{n},$$

then the deterministic equation is

$$\frac{dx}{dt} = -\alpha n x(1-x) + \beta(1-x).$$

Here the first term comes from the reaction $A + B \rightarrow 2B$, which decreases A , and the second term comes from $B \rightarrow A$, which increases A .

To compare the stochastic and deterministic descriptions, we simulated the stochastic model by the Gillespie algorithm for two system sizes, $n = 10$ and $n = 100$, using the same initial fraction $A(0)/n = 0.5$. For each value of n , several sample paths were generated and plotted together with the numerical solution of the deterministic ODE.

Figure 3 shows that the qualitative behavior is the same in both descriptions: the system tends toward the state with A/n close to 1. This agrees with the earlier observation that the state $A = n, B = 0$ is absorbing for the stochastic system.

For $n = 10$, the stochastic trajectories fluctuate quite a lot around the deterministic curve. Because the system is small, each individual reaction causes a noticeable jump, so randomness has a strong effect.

For $n = 100$, the stochastic paths are much closer to one another and follow the deterministic solution more closely. Thus, as the system size increases, the deterministic ODE gives a better approximation to the average behavior of the stochastic model.

In summary, the stochastic simulations confirm the same long-term trend predicted by the deterministic model, but the agreement is much better for larger n . Small systems show stronger noise, while large systems behave more like the smooth deterministic trajectory.

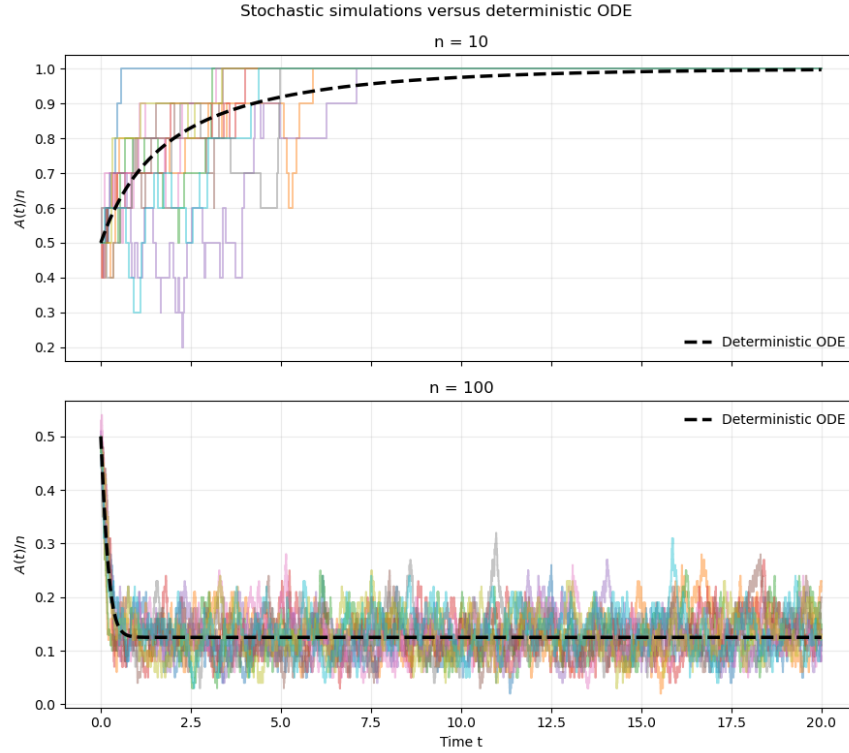


Figure 3: Comparison of stochastic simulations and the deterministic ODE for $n = 10$ and $n = 100$. The colored step curves are stochastic sample paths, and the dashed black curve is the deterministic solution.

Problem 4

Suppose you have expression data counts for a gene of interest. By looking at a tissue sample and modern technology, you were able to determine the number of mRNA expressed in 8 different cells during a 10 minute period of time. The protein counts are $y_1 = 120, y_2 = 150, y_3 = 100, y_4 = 90, y_5 = 160, y_6 = 130, y_7 = 110, y_8 = 140$. Assume that all cells have the same underlying baseline production level and that the system is modeled using a Poisson process.

- Suppose the prior distribution for the intensity θ of the Poisson process is distributed from a gamma distribution $\text{Gamma}(a, b)$, where $a = 2, b = 0.4$. Carry out a Bayesian algorithm to determine the posterior distributions using the determined data.
- Calculate the mean and the standard deviation of θ using the posterior distribution. *Note: think about what distribution θ has and use available*

formulas for such values.

- c) Consider abstract data $y_1, \dots, y_n$ for this same problem, where these numbers are drawn from a Gaussian distribution with mean μ and variance σ^2 . Calculate the expected value of θ for this data, and determine what happens to the variance as $n \rightarrow \infty$.

Let the observed counts be

$$y_1 = 120, y_2 = 150, y_3 = 100, y_4 = 90, y_5 = 160, y_6 = 130, y_7 = 110, y_8 = 140.$$

Assume that, conditional on the intensity θ , the data are independent Poisson random variables:

$$y_i \mid \theta \sim \text{Poisson}(\theta), \quad i = 1, \dots, 8.$$

We also assume a gamma prior

$$\theta \sim \Gamma(a, b),$$

with $a = 2$ and $b = 0.4$, where a is the shape parameter and b is the rate parameter.

Posterior distribution

For Poisson data, the likelihood is

$$L(\theta; y_1, \dots, y_8) = \prod_{i=1}^8 \frac{e^{-\theta} \theta^{y_i}}{y_i!} \propto \theta^{\sum_{i=1}^8 y_i} e^{-8\theta}.$$

By Bayes' rule,

$$\pi(\theta \mid y_1, \dots, y_8) \propto L(\theta; y_1, \dots, y_8) \pi(\theta).$$

Since the prior is $\Gamma(a, b)$, we have

$$\pi(\theta) \propto \theta^{a-1} e^{-b\theta}.$$

Therefore,

$$\pi(\theta \mid y_1, \dots, y_8) \propto \theta^{a-1+\sum y_i} e^{-(b+8)\theta}.$$

This is again a gamma distribution. Hence,

$$\theta \mid y_1, \dots, y_8 \sim \Gamma\left(a + \sum_{i=1}^8 y_i, b + 8\right).$$

Now

$$\sum_{i=1}^8 y_i = 1000,$$

so the posterior is

$$\theta \mid y_1, \dots, y_8 \sim \Gamma(1002, 8.4).$$

Posterior mean and standard deviation

For a gamma distribution $\Gamma(a, b)$ with shape a and rate b ,

$$E[\theta] = \frac{a}{b}, \quad \text{Var}(\theta) = \frac{a}{b^2}.$$

Thus, for the posterior,

$$E[\theta \mid y_1, \dots, y_8] = \frac{1002}{8.4} \approx 119.29,$$

and

$$\text{SD}(\theta \mid y_1, \dots, y_8) = \sqrt{\frac{1002}{8.4^2}} \approx 3.77.$$

Figure 4 shows the posterior density and indicates its center and spread obtained by carrying out a Bayesian algorithm.

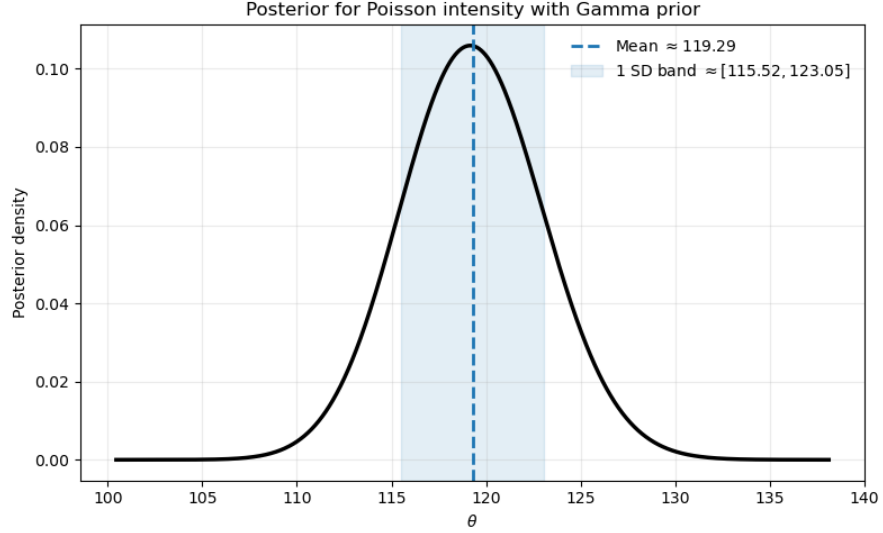


Figure 4: Posterior distribution for the Poisson intensity θ with a Gamma prior.

Gaussian data and asymptotic behavior

Now suppose instead that $y_1, \dots, y_n$ are drawn from a Gaussian distribution with mean μ and variance σ^2 , and we use the same idea of Bayesian updating for the unknown parameter θ . In the standard conjugate normal model, the posterior mean is a weighted average of the prior mean and the sample mean, and it converges to the sample mean as n becomes large. Thus the expected

value of θ is driven by the observed data, and in the large-sample limit,

$$E[\theta \mid y_1, \dots, y_n] \rightarrow \bar{y},$$

where $\bar{y}$ is the sample mean.

At the same time, the posterior variance shrinks as more data are observed. In particular, as $n \rightarrow \infty$,

$$\text{Var}(\theta \mid y_1, \dots, y_n) \rightarrow 0.$$

So the posterior becomes more concentrated around the true value suggested by the data.

Conclusion

Using a Gamma prior $\Gamma(2, 0.4)$ and the observed Poisson counts, the posterior is

$$\Gamma(1002, 8.4).$$

Its mean is approximately 119.29, and its standard deviation is approximately 3.77. For Gaussian data, the posterior mean approaches the sample mean as the number of observations grows, while the posterior variance tends to zero.